AI in Industry

Weekend 1, 16 and 17 October

Large language models

Tasks in words: answer, write, summarize, act

Project LLM marketing agent

Foundations

Language model A transformer gives each possible next word a chance; one is drawn, then the next.
Use it when the input and the output are text.
In industry chat assistants, drafting, summaries, translation.

Embedding A text as a row of numbers: similar meanings, similar rows.
Use it when comparing or searching texts by meaning.
In industry search by meaning; the search step of RAG.

Prompting and retrieval

Prompt engineering A brief with role, task, examples and format; chain or tree of thought for reasoning in steps.
Use it when starting any task: it trains nothing.
In industry customer replies, marketing texts, sorting messages.

RAG Retrieval-augmented generation: search your own documents, then answer from the pages found.
Use it when answers must rest on your documents and cite them.
In industry DoorDash's support bot answers couriers from its help articles (2024).

Prompt optimization

OPRO Optimization by prompting: an LLM reads earlier prompts with their scores and writes better ones.
Use it when you have a score and examples to compute it on.
In industry tuning a prompt to a metric, with no retraining.

GEPA An LLM reads why a prompt failed and rewrites it; every prompt that is best on some example is kept.
Use it when examples are few and each run is costly.
In industry Databricks: an open model with its prompts beat costlier models at extracting data from documents (2025).

Training

Supervised fine-tuning Train the model further on examples of good answers written by people.
Use it when a prompt cannot give the manner or format you need.
In industry chat assistants: InstructGPT (OpenAI, 2022).

Human feedback (RLHF) People rank answers, a scorer learns their taste, the model is nudged toward high scores.
Use it when a good answer is a matter of taste.
In industry ChatGPT (OpenAI, 2022).

Verifiable rewards (RLVR) A program checks each answer; right answers become more likely.
Use it when a program can check the answers: maths, code.
In industry reasoning models such as DeepSeek-R1 (2025).

Agents

Harness Software around the model: the model picks a tool, the harness runs it and returns the result, in a loop.
Use it when the task needs actions in files, databases or the web.
In industry coding agents, deep research, support bots that act.

Planning strategies Try possible steps in a search tree before acting: Monte Carlo tree search, LATS; UCB picks the branch.
Use it when a long task punishes an early wrong step.
In industry AlphaDev found faster sorting code by tree search, now in a standard C++ library (DeepMind, 2023).

Production

MLOps Machine learning operations: one pipeline from data to training, evaluation, deployment and monitoring.
Use it when a model goes live and must stay reproducible.
In industry Uber's Michelangelo, one platform for all its teams (2024).

Weekend 2, 13 and 14 November

Representation learning

Matching: which product for which customer

Project Recommender for online retail

Similarity

Inner product Multiply the matching numbers of two rows and add: one score. Cosine similarity: the same on rows scaled to length one.
Use it when scoring a customer against a product.
In industry the score inside search and recommenders.

Nearest neighbors Predict from the most similar customers or products.
Use it when you need a fast shortlist, or "customers like you".
In industry the first shortlist of search at Etsy (2024) and Airbnb (2025).

Learned profiles

Matrix product Every customer's row times every product's column: all the scores at once.
Use it when scoring every customer against every product.
In industry what GPUs compute fastest; inside recommenders and LLMs.

Matrix factorization A few learned numbers per customer and per product; their inner products fill in the missing ratings.
Use it when there are many customers and products, few ratings each.
In industry audiobook recommendations at Spotify, with the numbers learned by networks (2024).

Gradient descent Change every number a small step that lowers the error; repeat.
Use it when a model has many numbers to learn.
In industry trains recommenders, neural networks and LLMs.

Bandits

Multi-armed bandit Show one option, learn from the click, choose again: explore or exploit.
Use it when choices repeat, feedback is fast and the best option is unknown.
In industry which offer, page or content to show.

Random Pick an option at random each time.
Use it when you need a baseline, or a fair test.
In industry the random split of visitors in an A/B test.

Explore then exploit Try every option for a while, then keep the best.
Use it when the best option does not change.
In industry an A/B test, then the winner for everyone.

Epsilon-greedy The best option so far; with chance epsilon, a random one.
Use it when you want a simple rule that never stops exploring.
In industry the mix of music and podcasts on Spotify's home page (2025).

UCB Upper confidence bound: the highest estimate plus a bonus for uncertainty.
Use it when every try is costly.
In industry Amazon tested it on the application pages of Amazon Payments (2026).

Scale

Parallelized matrix factorization Split customers, products and ratings over many computers that work at once.
Use it when the data or the model do not fit one computer.
In industry LinkedIn trains its ranking models on many GPUs at once (2024).

Computing clusters Thousands of GPUs joined by fast links; a run costs GPU hours, energy and money.
Use it when training or serving large models.
In industry Apertus, the Swiss open LLM, trained on Alps at CSCS (2025).

Weekend 3, 11 and 12 December

Reinforcement learning

Decisions in sequence, rewards later

Project RL vehicle routing

Problem

Markov decision process States, actions, rewards and the chances of the next state; a policy says what to do in each state.
Use it when each decision changes the next situation.
In industry maintenance, pricing, stock, delivery routes, robots.

Exploration

Epsilon-greedy The bandit rule of weekend 2, used in every state.
Use it when you learn while you act.

Values

Bellman equation The value of an action: its reward now plus the value of the best next action.
Use it when the rules and their chances are known.
In industry planning maintenance and stock with known odds.

Robbins-Monro Learn an average from noisy samples: move the guess a small step toward each one.
Use it when every observation is noisy.
In industry the update inside SARSA, Q-learning and stochastic gradient descent.

Learning from experience

SARSA and Q-learning Learn the value of each action from your own steps, with no model of the world.
Use it when there is no model, only experience or a simulator.
In industry Lyft matches riders and drivers by future earnings learned from live trips (2024).

REINFORCE Make the actions of well-paid episodes more likely: learn the policy itself.
Use it when actions are many, or the policy is the goal.
In industry the family that trains chat and reasoning models of weekend 1.